# Nose & nasal cavity

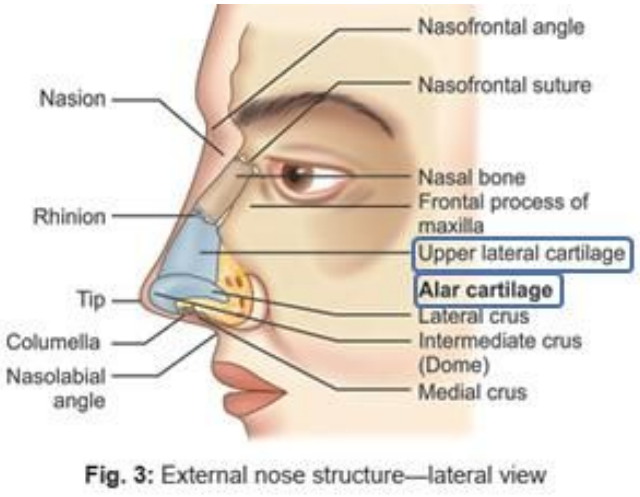
External framework, the four walls of each cavity, septum, osteomeatal complex, mucosa, vessels and nerves.


1. External nose
2. Danger area of the face
3. Nasal cavity: walls and regions
4. Turbinates and meatuses
5. Nasal septum
6. Middle meatus and osteomeatal complex
7. Lining of the nose
8. Blood supply
9. Nerve supply and lymphatics
10. In theatre
11. On imaging

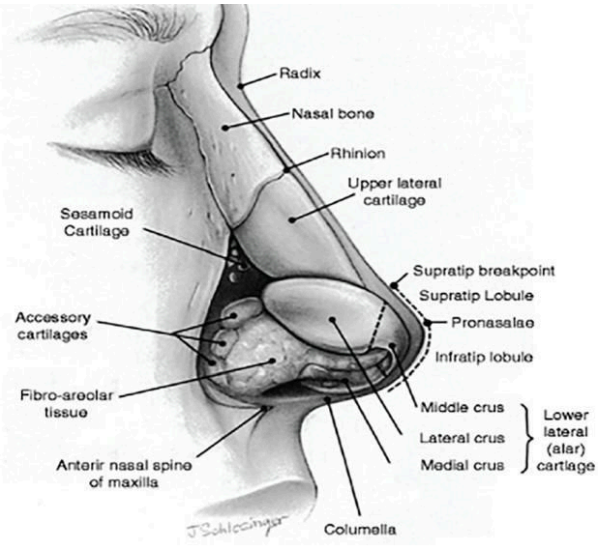

## 01 External nose

- Pyramid of an **osteocartilaginous framework**: upper third bony (paired **nasal bones** on the frontal bone, between the frontal processes of the maxillae); lower two-thirds cartilaginous.
- **Upper lateral cartilages:** under the nasal bones above; their lower free edge forms the **internal nasal valve**. They fuse with the upper border of the septal cartilage.
- **Lower lateral (alar) cartilages:** U-shaped with a **lateral crus** (ala), **medial crus** (columella) and intermediate crus (**dome**, tip). The free alar rim is fibrofatty, not cartilage.
- Muscles: procerus, nasalis (transverse and alar parts), levator labii superioris alaeque nasi, depressor septi; all **facial nerve**.
- Skin is **thin and mobile over the dorsum**, **thick, adherent and sebaceous over the tip** (rhinophyma).

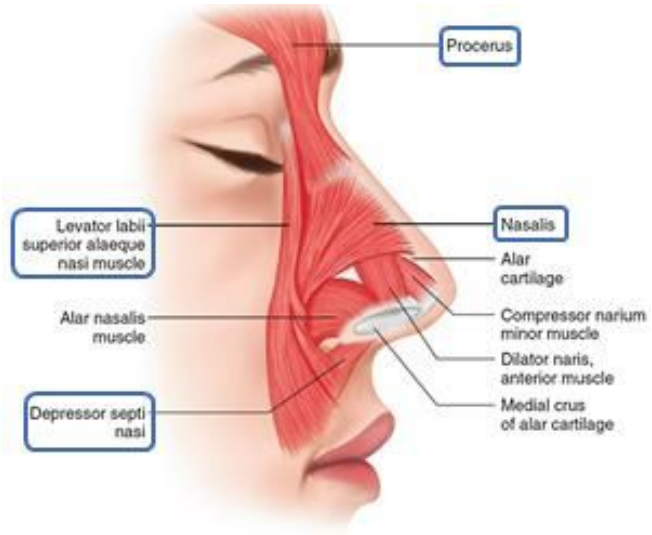
> **CLINICAL PEARL** Saddle (supratip) deformity follows loss of septal cartilage support: septal abscess, over-resection at SMR, trauma or granulomatous disease.



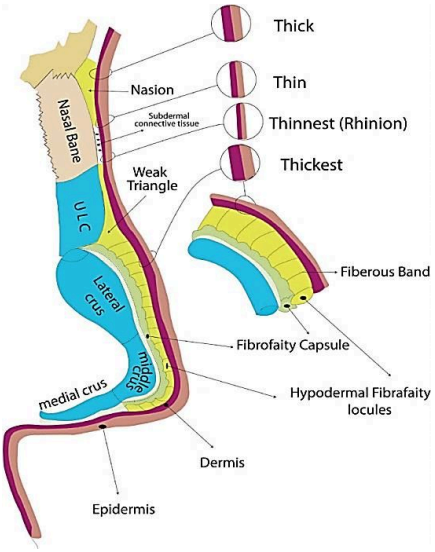

External nose: surface landmarks and framework, lateral view
ENT Steps Anatomy, p. 92



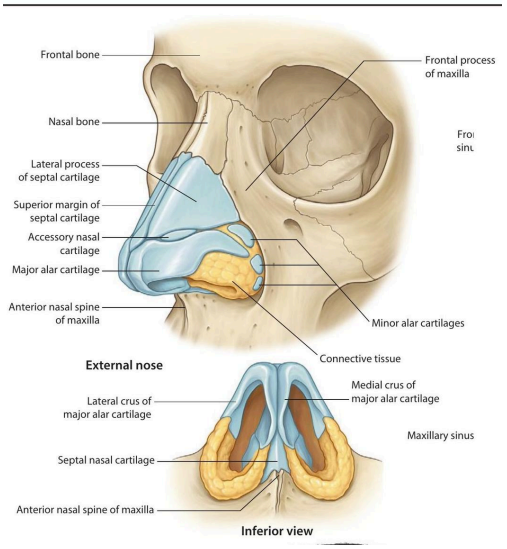

Osteocartilaginous framework of the external nose
ENT Steps Anatomy, p. 93




Muscles of the nose
ENT Steps Anatomy, p. 95




Nasal skin thickness from nasion to tip
ENT Steps Anatomy, p. 96

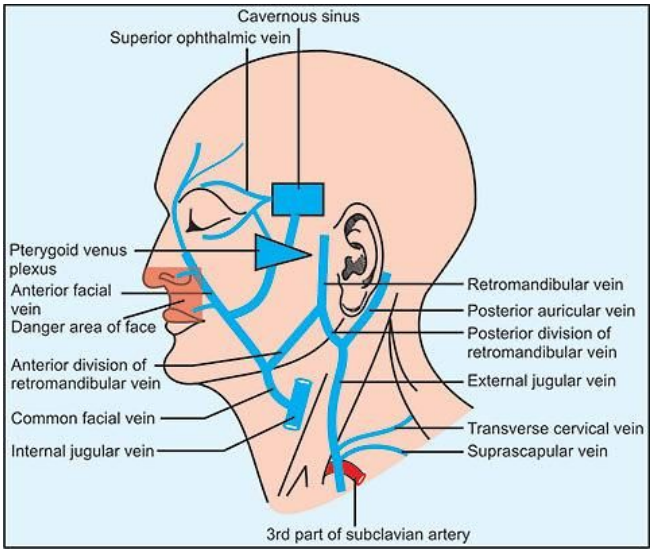



External nose: cartilages (lateral and inferior views)
Gray's Atlas of Anatomy, 2nd ed., p. 566

## 02 Danger area of the face

- Triangle from the root of the nose to the corners of the mouth.
- Facial veins here are **valveless** and connect to the **cavernous sinus** through the ophthalmic veins and the pterygoid plexus (deep facial vein).

> **CLINICAL PEARL** Squeezing a furuncle of the nasal vestibule or upper lip can cause cavernous sinus thrombosis.



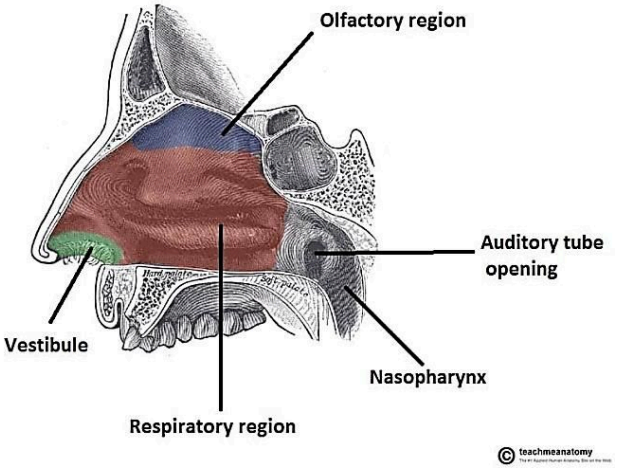

Danger area of the face: venous drainage to the cavernous sinus

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## 03 Nasal cavity: walls and regions

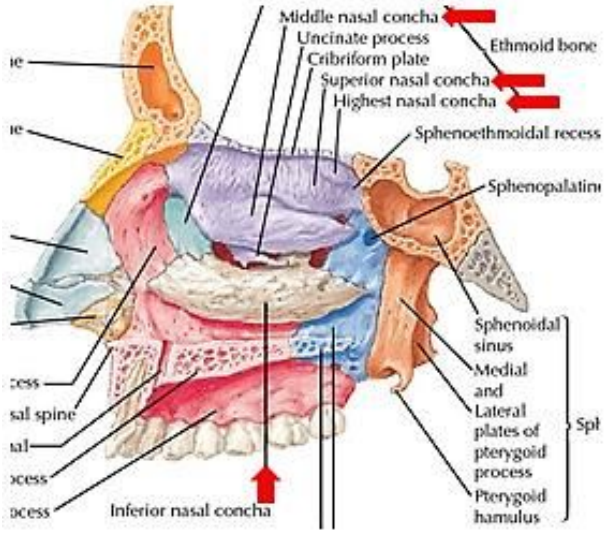
- Each cavity runs from the **nostril** to the **choana**. The skin-lined **vestibule** lies in front; the mucosa-lined cavity proper behind.
- **Roof:** sloping nasal bones in front, **cribriform plate** in the middle (olfactory nerves), body of sphenoid behind.
- **Floor:** palatine process of the maxilla (front three-quarters) and horizontal plate of the palatine bone (back quarter).
- **Medial wall:** the septum.
- **Lateral wall bones:** nasal, frontal process and medial surface of maxilla, lacrimal, ethmoid labyrinth, inferior turbinate, perpendicular plate of palatine, medial pterygoid plate.
- **Internal nasal valve:** the narrowest part of the nasal airway, bounded by the caudal edge of the upper lateral cartilage (limen nasi), the septum, the head of the inferior turbinate and the floor. Normal angle 10–15°.

> **CLINICAL PEARL** Cottle's test: pulling the cheek laterally improves breathing when the valve is the cause of obstruction.



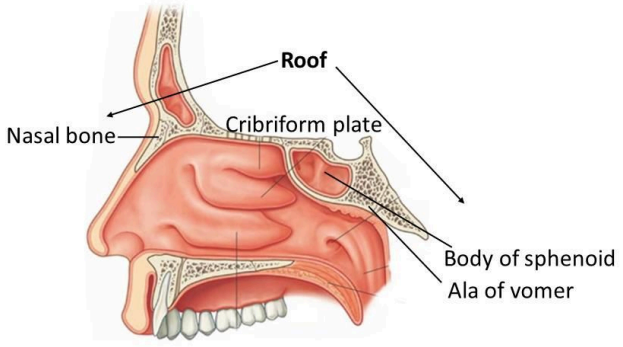

Regions of the nasal cavity: vestibule, respiratory and olfactory

ENT Steps Anatomy, p. 97



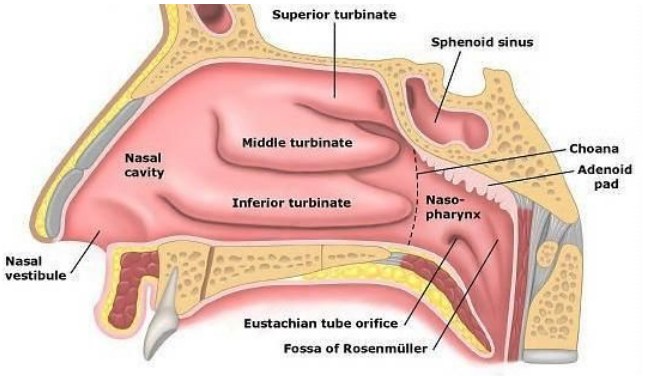

Lateral nasal wall: bones, turbinates and meatuses

ENT Steps Anatomy, p. 99



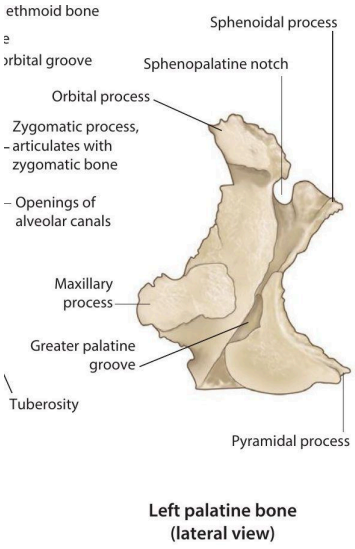

Roof of the nasal cavity: nasal bone, cribriform plate and sphenoid

Dr. Sameh Doss handout, p. 102



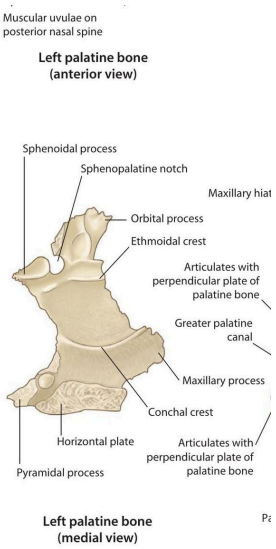

Lateral nasal wall with turbinates

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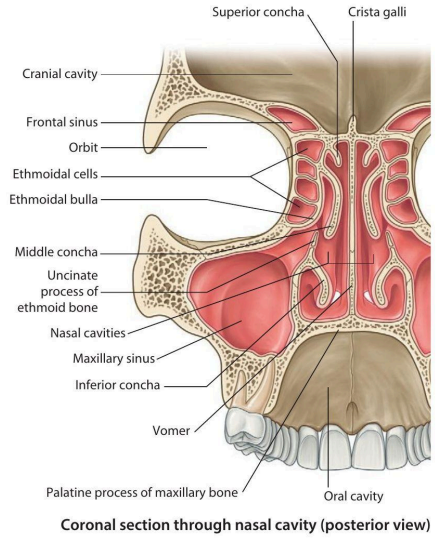

Left palatine bone (lateral view)

Gray's Atlas of Anatomy, 2nd ed., p. 487



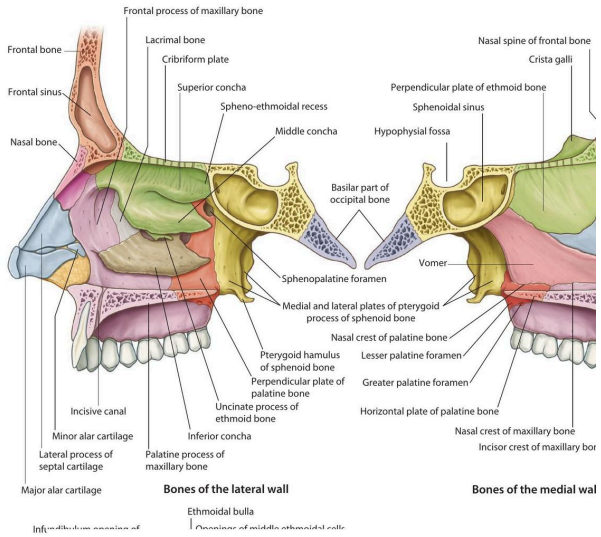

Left palatine bone (medial and posterior views)

Gray's Atlas of Anatomy, 2nd ed., p. 487



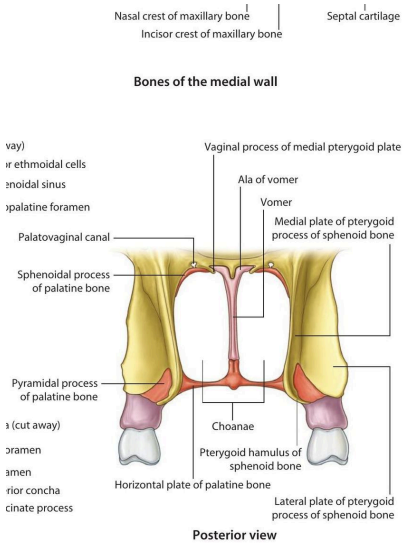

Coronal section through the nasal cavity (posterior view)

Gray's Atlas of Anatomy, 2nd ed., p. 566




Bones of the lateral and medial walls of the nasal cavity

Gray's Atlas of Anatomy, 2nd ed., p. 567




Nasal cavity, posterior view: choanae and pterygoid plates

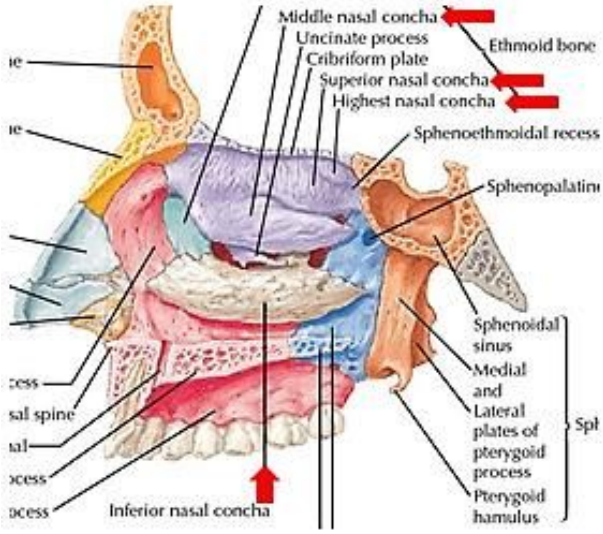
Gray's Atlas of Anatomy, 2nd ed., p. 567

## 04 Turbinates and meatuses

- **Inferior turbinate:** largest, a **separate bone**. **Middle, superior and supreme:** parts of the ethmoid.
- **Inferior meatus: nasolacrimal duct** opens in its anterior part, about 1 cm behind the front of the turbinate, guarded by **Hasner's valve**.
- **Middle meatus:** the **osteomeatal complex**: frontal, maxillary and anterior ethmoid sinuses drain here.
- **Superior meatus:** posterior ethmoid cells.
- **Sphenoethmoidal recess:** above the superior turbinate; the **sphenoid ostium** opens here, about 1–1.5 cm above the upper margin of the choana, close to the septum.

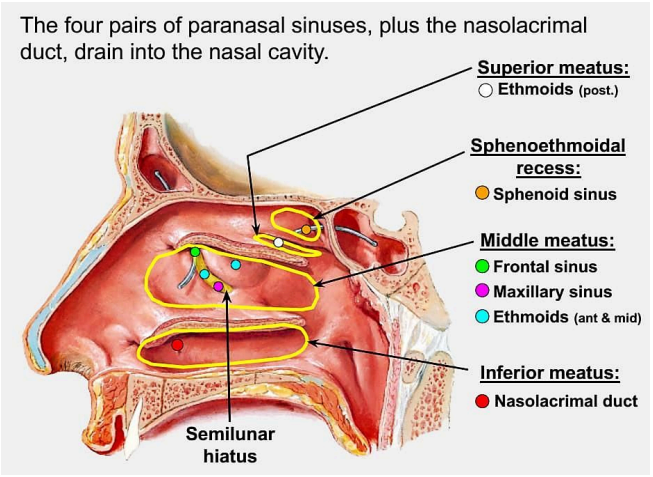
**CLINICAL PEARL** Sinuses that drain in front of the basal lamella (anterior group) send mucus along the lateral pharyngeal wall; posterior group mucus runs over the posterior pharyngeal wall.

**SOURCES DIFFER SPHENOID OSTIUM HEIGHT:** ENT Steps gives about 1 cm above the choana in the nose chapter and 1.5 cm in the sinus chapter.



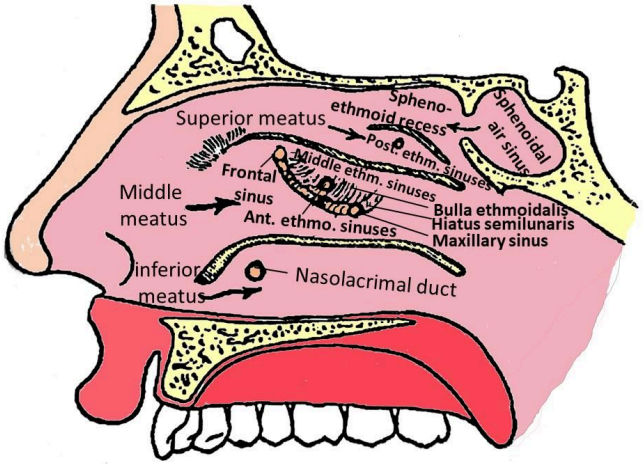

Lateral nasal wall: bones, turbinates and meatuses

ENT Steps Anatomy, p. 99



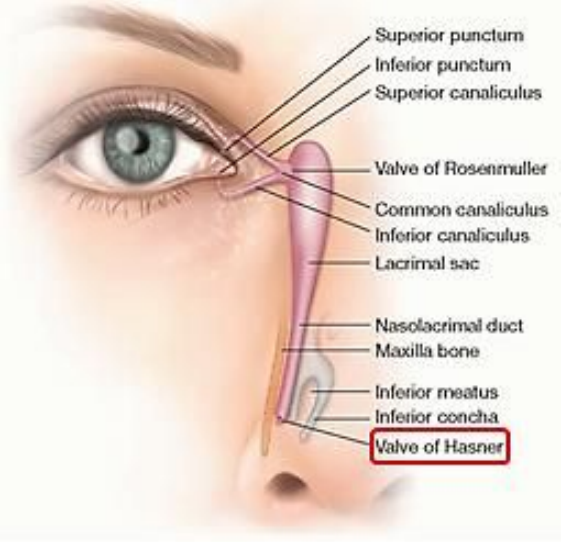

Drainage of the sinuses and nasolacrimal duct into the meatuses

ENT Steps Anatomy, p. 101



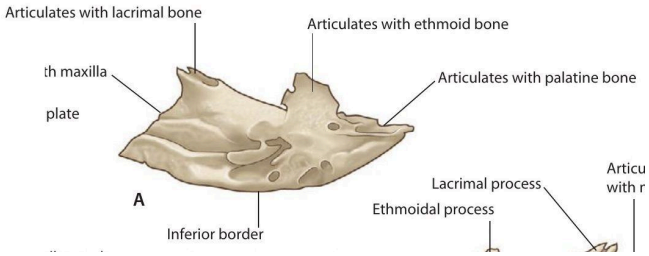

Lateral wall of the nasal cavity

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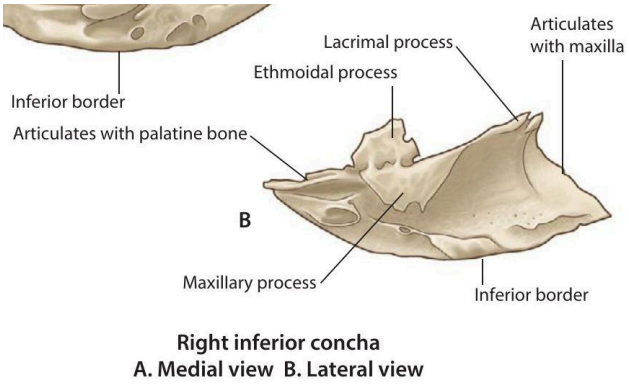

Lacrimal drainage: canaliculi, sac, nasolacrimal duct and Hasner's valve

ENT Steps Anatomy, p. 100




Right inferior concha

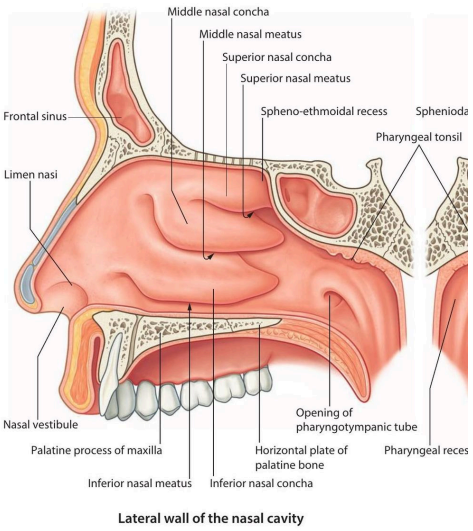
Gray's Atlas of Anatomy, 2nd ed., p. 486




**Right inferior concha**
**A. Medial view B. Lateral view**

Right inferior concha: medial and lateral views

Gray's Atlas of Anatomy, 2nd ed., p. 486



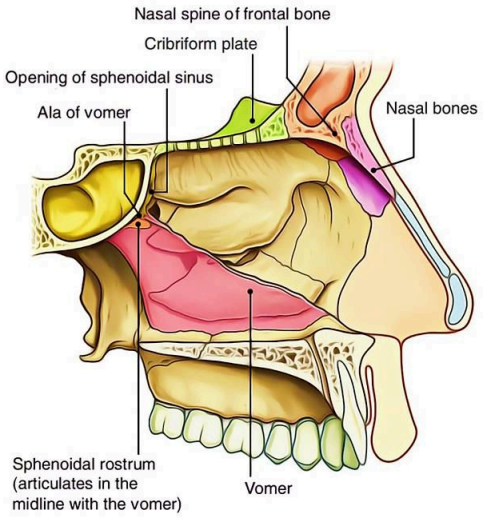

Lateral wall of the nasal cavity: conchae and meatuses

Gray's Atlas of Anatomy, 2nd ed., p. 568

## 05 Nasal septum

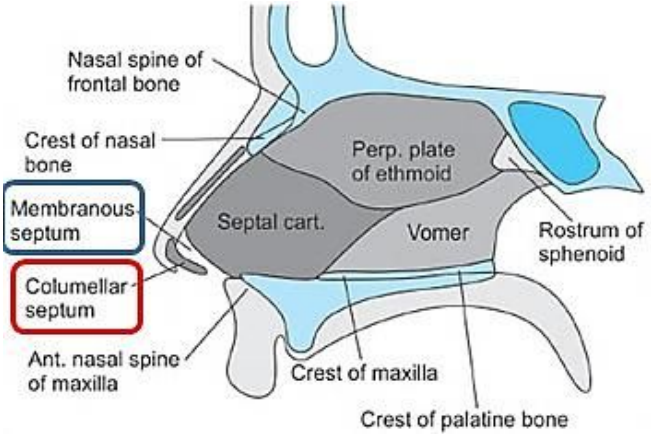
- Three parts: **columellar septum** (medial crura of alar cartilages), **membranous septum** (double layer of skin only) and **septum proper**.
- **Septum proper: quadrilateral septal cartilage**, perpendicular plate of the ethmoid, and the **vomer**. Small contributions: nasal spine of frontal, crests of nasal, maxillary and palatine bones, anterior nasal spine, rostrum of sphenoid.
- The cartilage's lower edge sits in the vomerine groove and on the anterior nasal spine. Dislocation from the spine gives **caudal deviation**; from the vomerine groove gives a **septal spur**.

> **CLINICAL PEARL** Preserve an L-strut of at least 1 cm dorsal and caudal cartilage at septoplasty to prevent saddle nose and tip collapse.



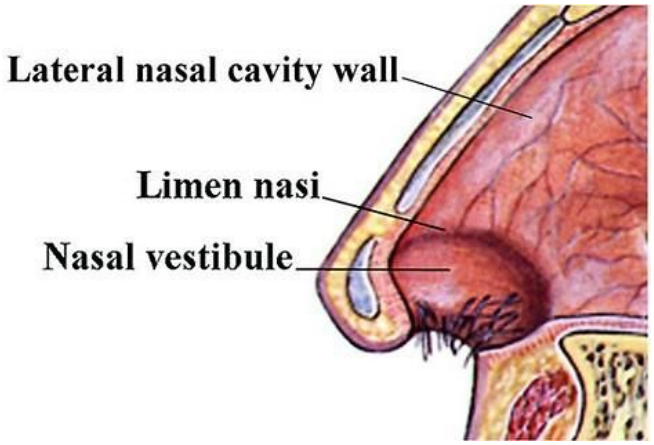

Bones of the nasal septum

ENT Steps Anatomy, p. 102

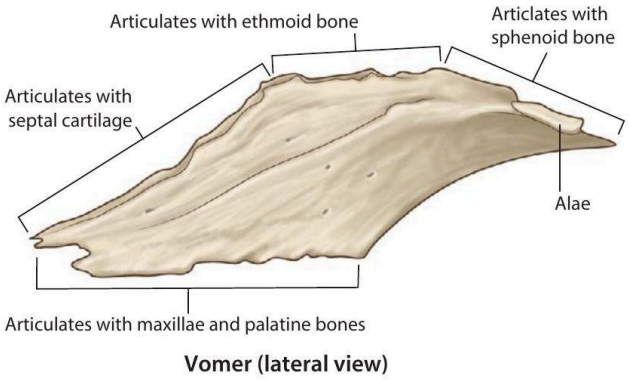



Nasal septum: cartilage and bones

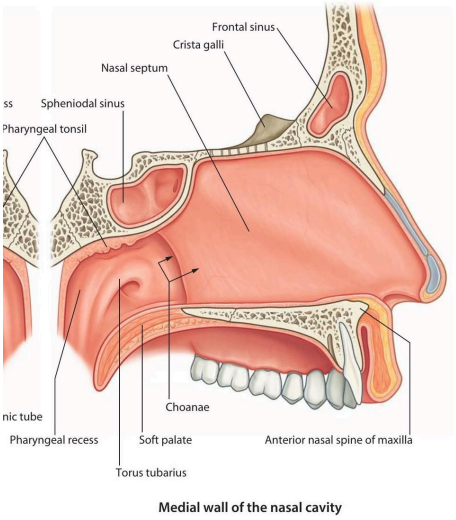
ENT Steps Anatomy, p. 104




Limen nasi and nasal vestibule
ENT Steps Anatomy, p. 103




Vomer (lateral view)
Gray's Atlas of Anatomy, 2nd ed., p. 486

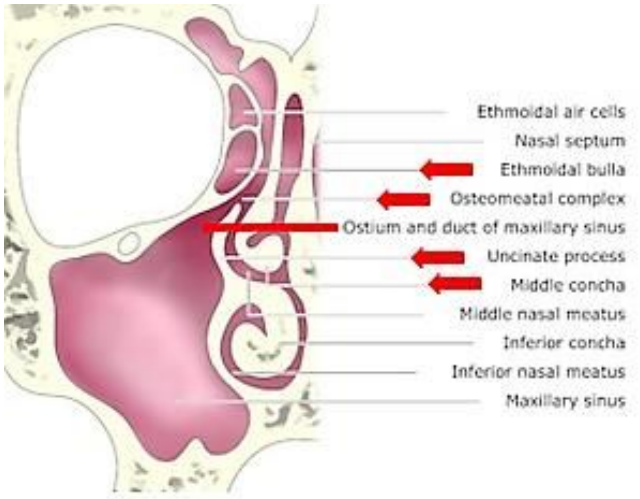



Medial wall of the nasal cavity: septum
Gray's Atlas of Anatomy, 2nd ed., p. 568

## 06 Middle meatus and osteomeatal complex

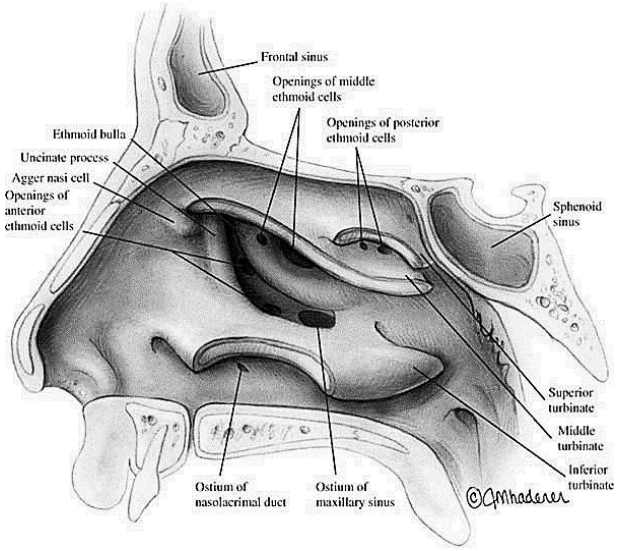
- **OMC** = middle turbinate, uncinate process, bulla ethmoidalis and ethmoidal infundibulum, with the ostia of the frontal, maxillary and anterior ethmoid sinuses.
- **Middle turbinate basal (ground) lamella:** anterior third vertical to the **lateral edge of the cribriform plate**; middle third frontal to the **lamina papyracea**; posterior third horizontal to the lateral nasal wall.
- **Uncinate process:** sickle-shaped ethmoid lamina; free posterosuperior edge parallel to the bulla. Upper end inserts on the **lamina papyracea, skull base or middle turbinate**, which decides where the frontal sinus drains.
- **Hiatus semilunaris:** 2D gap between the uncinate and the bulla, leading into the **ethmoidal infundibulum** (3D space).
- **Bulla ethmoidalis:** largest anterior ethmoid cell; spaces above and behind it form the **lateral sinus (of Grünwald)**.
- **Fontanelles:** bone-free areas above and below the uncinate's lower end; an accessory maxillary ostium (about 30%) opens here.
- **Paradoxical middle turbinate** curves laterally; **concha bullosa** is a pneumatised middle turbinate; both can narrow the middle meatus.

**CLINICAL PEARL** Mucus ignores an accessory ostium and returns to the natural ostium; failing to join the two at antrostomy causes recirculation.



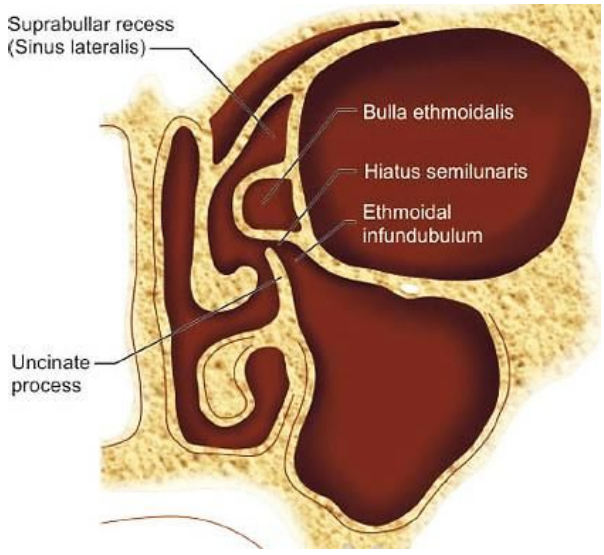

Osteomeatal complex in coronal section

ENT Steps Anatomy, p. 108



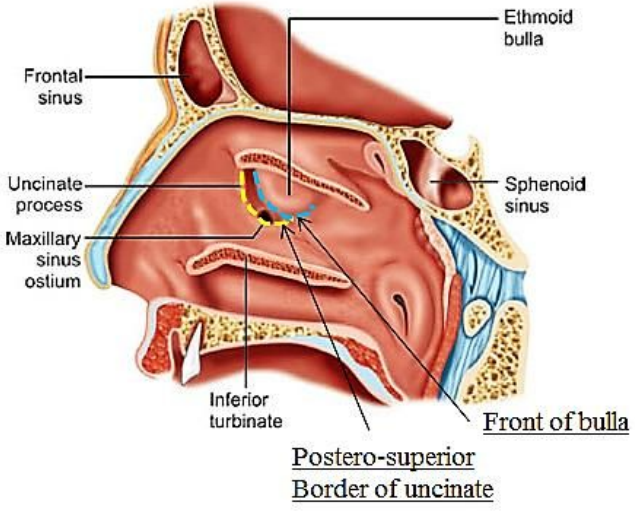

Lateral wall with the middle turbinate removed: uncinate, bulla, hiatus semilunaris

ENT Steps Anatomy, p. 108



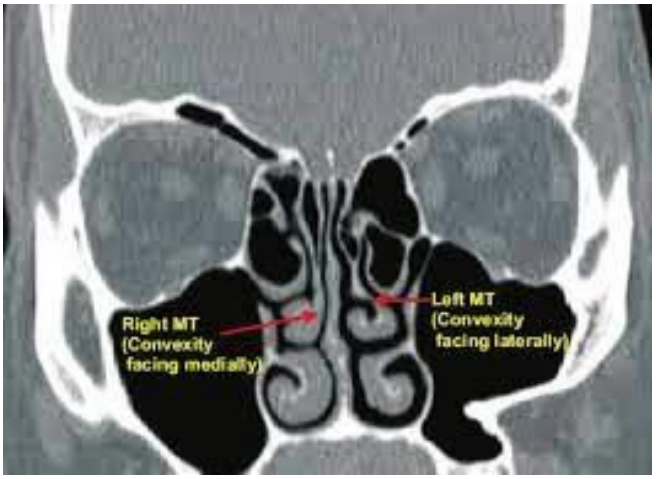

Ethmoidal infundibulum and hiatus semilunaris

ENT Steps Anatomy, p. 112



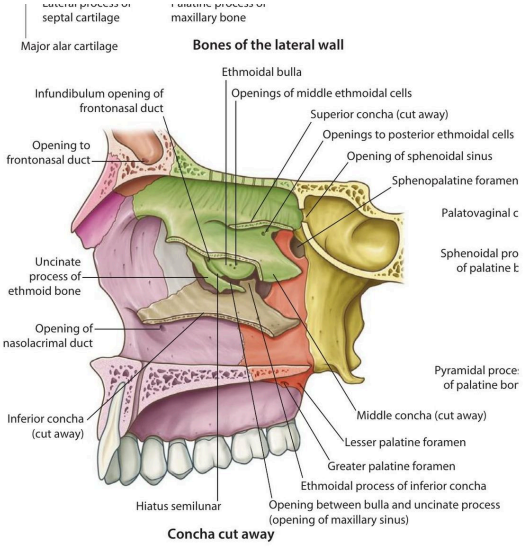

Lateral nasal wall structures around the middle meatus

ENT Steps Anatomy, p. 113



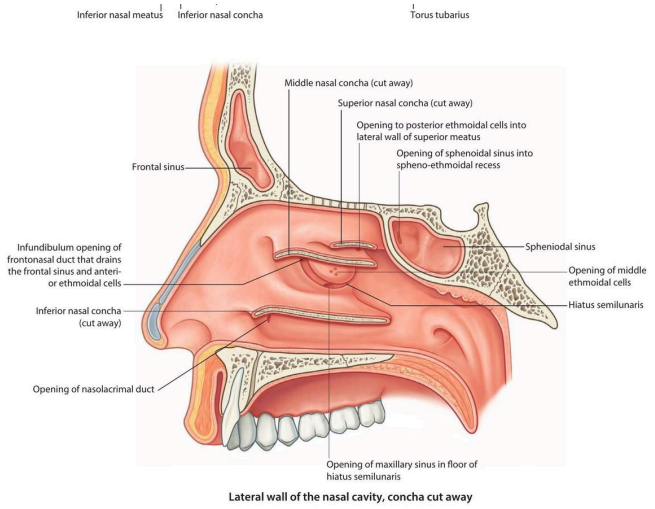

Coronal CT: paradoxical middle turbinate

ENT Steps Anatomy, p. 109




Lateral nasal wall, concha cut away: openings of the sinuses

Gray's Atlas of Anatomy, 2nd ed., p. 567

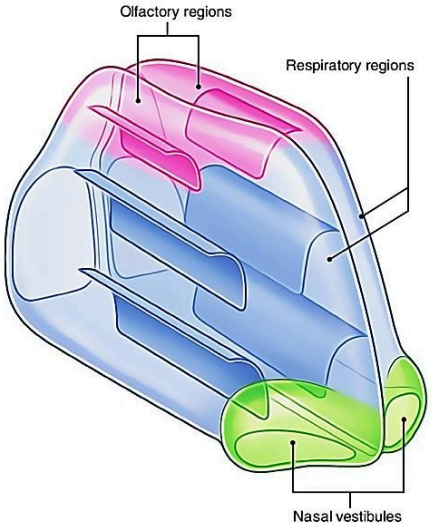



Lateral wall of the nasal cavity, conchae cut away

Gray's Atlas of Anatomy, 2nd ed., p. 568

## 07 Lining of the nose

- **Vestibule:** stratified squamous skin with vibrissae and sebaceous glands.
- **Olfactory region:** roof, upper septum and above the superior turbinate; paler.
- **Respiratory mucosa:** pseudostratified ciliated columnar with goblet cells, seromucous glands and **erectile venous sinusoids** (thickest on the turbinates).
- Sympathetic stimulation constricts (decongests); parasympathetic dilates and produces watery secretions.




Mucosal regions of the nasal cavity

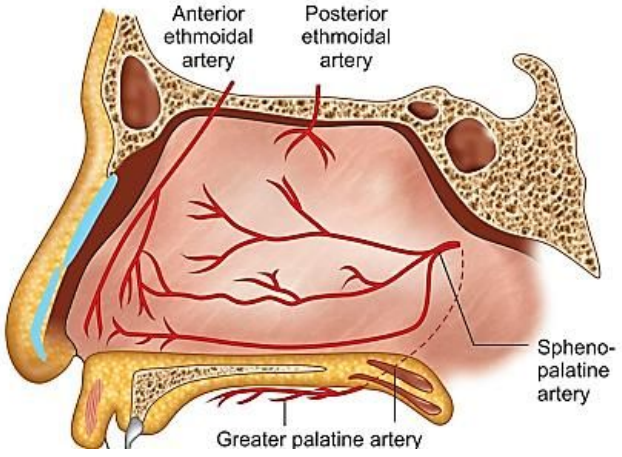
ENT Steps Anatomy, p. 117

## 08 Blood supply

- **ICA via ophthalmic:** anterior and posterior ethmoidal arteries (upper part).
- **ECA: sphenopalatine** (main supply, through the sphenopalatine foramen behind the middle turbinate), greater palatine, superior labial (facial).
- **Little's area (Kiesselbach's plexus)** on the anterior septum: anterior ethmoidal, sphenopalatine (septal branch), greater palatine and superior labial arteries. Most common site of epistaxis.
- **Woodruff's plexus:** posterior inferior lateral wall below the inferior turbinate; source of posterior bleeds in older patients.

**CLINICAL PEARL** Refractory posterior epistaxis: endoscopic sphenopalatine artery ligation. Bleeding from above the middle turbinate suggests the ethmoidal arteries.

**SOURCES DIFFER WOODRUFF'S PLEXUS:** Dhingra and Bansal describe it as venous; endoscopic studies show it is mainly arterial (sphenopalatine branches).




Blood supply of the lateral nasal wall

ENT Steps Anatomy, p. 118

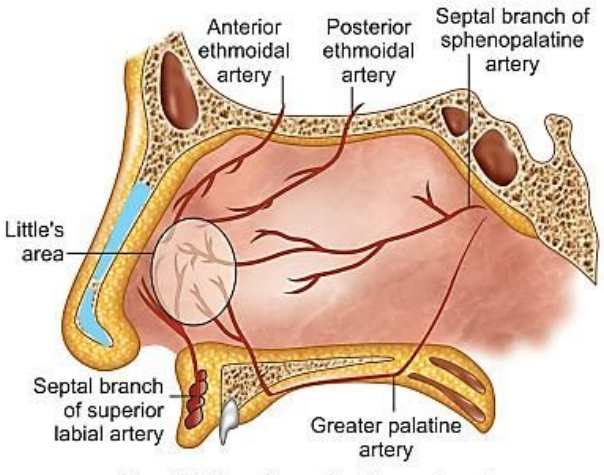
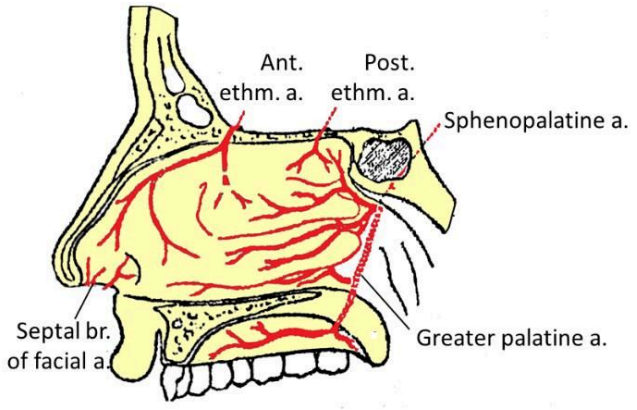



**Fig. 15:** Blood supply of nasal septum

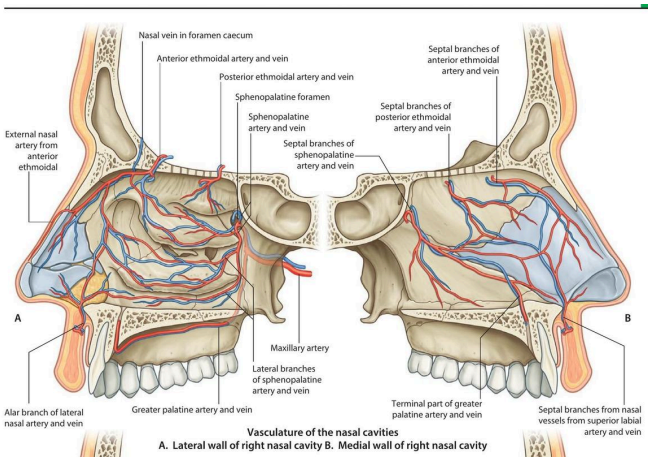
Blood supply of the nasal septum and Little's area

ENT Steps Anatomy, p. 118




Arterial supply of the nasal cavity

Dr. Sameh Doss handout, p. 104



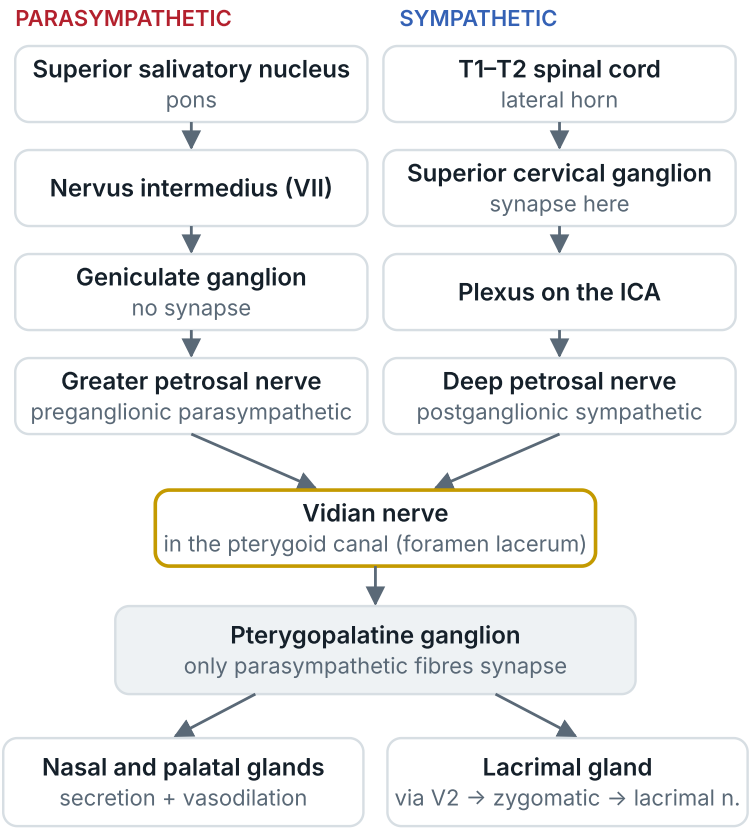

Vasculature of the nasal cavities: lateral and medial walls

Gray's Atlas of Anatomy, 2nd ed., p. 569

## 09 Nerve supply and lymphatics

- **Olfactory nerves:** 12–20 bundles through the cribriform plate to the olfactory bulb.
- **General sensation:** anterior ethmoidal nerve (V1) to the front and upper part; branches of the pterygopalatine ganglion (V2) to the back two-thirds; infraorbital to the vestibule.
- Block points for anaesthesia: **sphenopalatine foramen** (posterior end of middle turbinate) and **anterior ethmoidal nerve** high under the nasal bones.
- **Vidian nerve** = greater petrosal (parasympathetic from VII) + deep petrosal (sympathetic from the ICA plexus) → pterygopalatine ganglion. Only the parasympathetic fibres relay.
- **Lymphatics:** front of the nose → submandibular nodes; back → retropharyngeal and upper deep cervical nodes. Upper lymphatics connect to the subarachnoid space along the olfactory nerves.

> **CLINICAL PEARL** Vidian neurectomy reduces watery rhinorrhoea in vasomotor rhinitis but also dries the eye on that side (lacrimal fibres travel the same way).




Sympathetic fibres pass through unrelayed → vasoconstriction (decongestion

Where the two autonomic streams meet. Cutting the vidian nerve reduces watery rhinorrhoea but also tear secretion on that side.

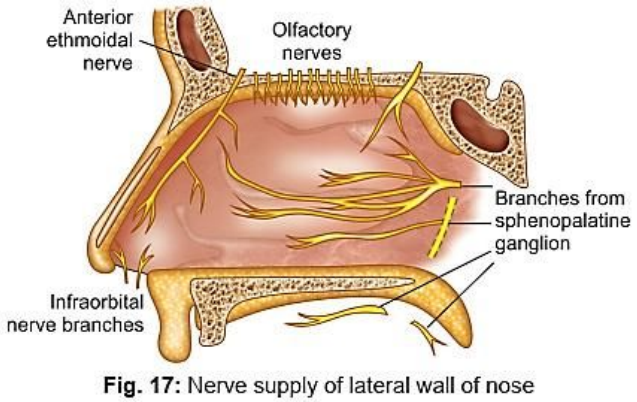
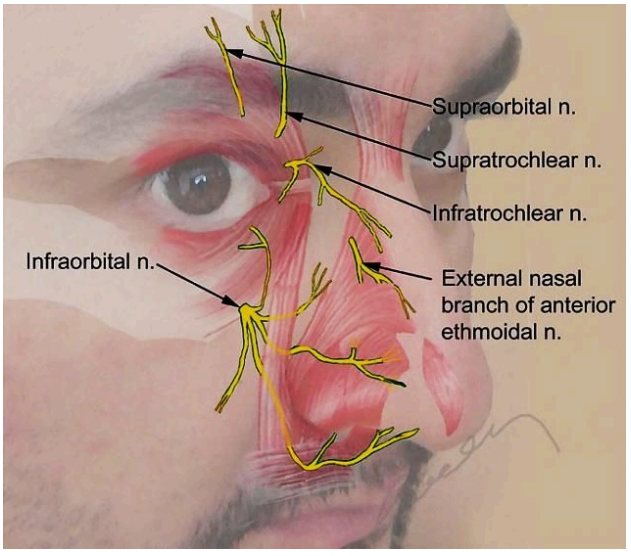



**Fig. 17:** Nerve supply of lateral wall of nose

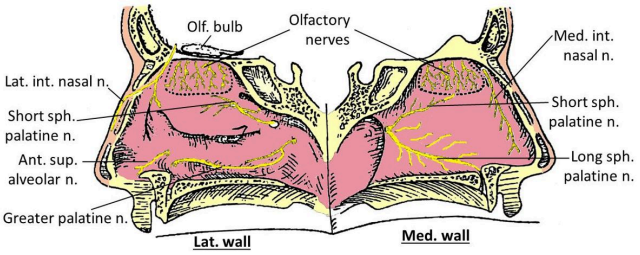
Nerve supply of the lateral nasal wall

ENT Steps Anatomy, p. 121



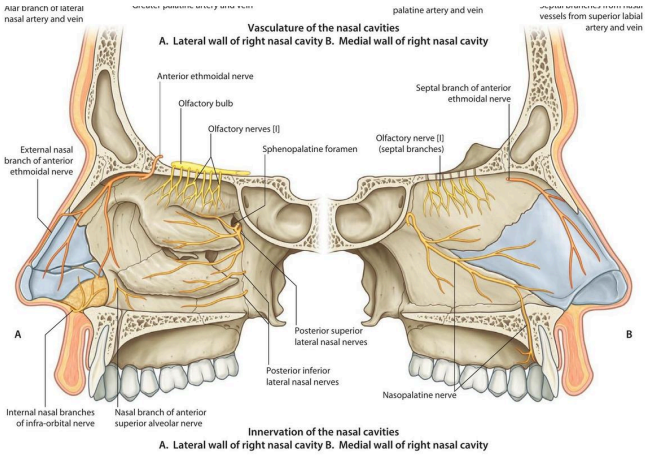

Sensory nerves of the external nose

ENT Steps Anatomy, p. 121




Nerve supply of the nasal cavity

Dr. Sameh Doss handout, p. 104



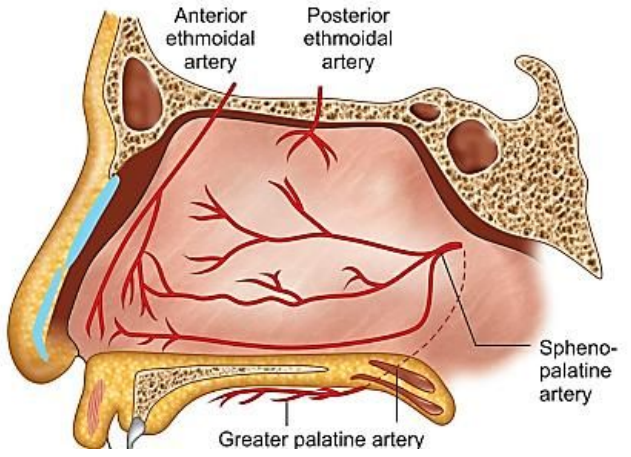

Innervation of the nasal cavities: lateral and medial walls

Gray's Atlas of Anatomy, 2nd ed., p. 569

## 10 Surgical anatomy: In theatre

- **Septoplasty:** Killian's incision 1 cm behind the caudal edge, or hemitransfixion for caudal deviations; elevate the mucoperichondrial flap; keep a 1 cm dorsal and caudal L-strut.
- **Inferior turbinate reduction:** submucosal, keeping mucosa; avoid the lateral wall where the nasolacrimal duct opens.
- **Sphenopalatine artery ligation:** raise a mucosal flap behind the middle turbinate's posterior attachment; the **crista ethmoidalis** marks the foramen; the artery often branches before exiting.
- **Anterior ethmoidal artery** ligation is done externally (Lynch incision) or endoscopically at the ethmoid roof.

> **CLINICAL PEARL** Septal haematoma must be drained urgently: the cartilage depends on the perichondrium for its blood supply and can necrose into a saddle nose.

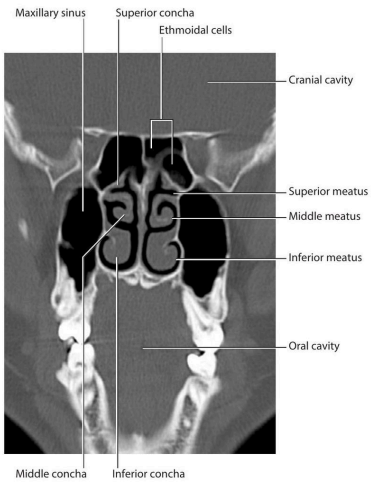



Blood supply of the lateral nasal wall

ENT Steps Anatomy, p. 118

## 11 Radiological anatomy: On imaging

- **CT coronal** shows the septum, turbinates, concha bullosa and the osteomeatal complex. **MRI** separates tumour from retained secretions.
- Nasal bone fractures are a clinical diagnosis; X-rays add little.




Coronal CT: nasal cavity, conchae and meatuses

Gray's Atlas of Anatomy, 2nd ed., p. 570

## Tables

### Blood supply of the nose

| Artery | From | Area |
|---|---|---|
| **Anterior ethmoidal** | Ophthalmic (ICA) | Anterosuperior septum and lateral wall |
| **Posterior ethmoidal** | Ophthalmic (ICA) | Superior turbinate area |
| **Sphenopalatine** | Maxillary (ECA) | Most of the lateral wall and septum (main supply) |
| **Greater palatine** | Maxillary (ECA) | Anteroinferior septum via incisive canal |
| **Superior labial** | Facial (ECA) | Vestibule and anterior septum |

Little's area = anterior ethmoidal + sphenopalatine + greater palatine + superior labial.

### Branches of the maxillary artery

| Part | Position | Branches |
|---|---|---|
| **1st (mandibular)** | Behind the neck of mandible | Deep auricular, anterior tympanic, middle meningeal, accessory meningeal, inferior alveolar |
| **2nd (pterygoid)** | On or deep to lateral pterygoid | Masseteric, deep temporal, pterygoid, buccal |
| **3rd (pterygopalatine)** | In the pterygopalatine fossa | Posterior superior alveolar, infraorbital, descending palatine, artery of pterygoid canal, pharyngeal, sphenopalatine |

### Parasympathetic ganglia of the head

| Ganglion | Preganglionic nerve | Hangs from | Supplies |
|---|---|---|---|
| **Ciliary** | Oculomotor (III) | Nasociliary (V1) | Sphincter pupillae, ciliary muscle |
| **Pterygopalatine** | Greater petrosal (VII) → vidian | Maxillary (V2) | Lacrimal, nasal, palatal and pharyngeal glands |
| **Otic** | Lesser petrosal (IX) | Mandibular (V3) | Parotid (via auriculotemporal) |
| **Submandibular** | Chorda tympani (VII) → lingual | Lingual (V3) | Submandibular and sublingual glands |

Only the parasympathetic fibres synapse; sympathetic and sensory fibres pass through.

## M Mnemonics

Little's area arteries

**LEGS**

Labial (superior), Ethmoidal (anterior), Greater palatine, Sphenopalatine

Septum bones

**Very Pretty Cartilage**

Vomer, Perpendicular plate of ethmoid, quadrilateral Cartilage

## Q Rapid recall

1. **What is the narrowest part of the nasal airway?** The internal nasal valve (bounded by the limen nasi, septum, inferior turbinate head and floor).
2. **Which sinuses drain into the middle meatus?** Frontal, maxillary and anterior ethmoid.
3. **Where does the nasolacrimal duct open?** Anterior part of the inferior meatus, guarded by Hasner's valve.
4. **Four arteries forming Kiesselbach's plexus?** Anterior ethmoidal, sphenopalatine (septal branch), greater palatine and superior labial.
5. **Main components of the bony and cartilaginous septum?** Quadrilateral cartilage, perpendicular plate of ethmoid and vomer.
6. **Three attachments of the middle turbinate basal lamella?** Anterior third to the cribriform plate, middle third to the lamina papyracea, posterior third to the lateral nasal wall.
7. **What forms the vidian nerve?** Greater petrosal (parasympathetic) plus deep petrosal (sympathetic).
8. **What is the hiatus semilunaris?** The 2D gap between the free edge of the uncinate process and the bulla ethmoidalis, leading into the ethmoidal infundibulum.
9. **What forms the upper third of the external nose?** The paired nasal bones.
10. **What are the three crura of the alar cartilage?** Lateral, intermediate (dome) and medial.
11. **Normal internal nasal valve angle?** About 10–15°.
12. **What is Cottle's test?** Pulling the cheek laterally; improved breathing suggests valve obstruction.
13. **What forms the floor of the nasal cavity?** Palatine process of the maxilla (front 3/4) and horizontal plate of palatine (back 1/4).
14. **Which turbinate is a separate bone?** The inferior turbinate.
15. **What is the sphenoethmoidal recess?** Space above the superior turbinate into which the sphenoid sinus drains.
16. **Upper end of the uncinate process: three possible insertions?** Lamina papyracea, skull base or middle turbinate.
17. **What is Woodruff's plexus?** Venous plexus at the posterior lateral wall below the inferior turbinate; source of posterior epistaxis.
18. **Landmark for the sphenopalatine foramen?** Crista ethmoidalis, behind the posterior attachment of the middle turbinate.
19. **Lymph drainage of the anterior nose?** Submandibular nodes.
20. **Which nerve supplies the nasal tip skin?** External nasal branch of the anterior ethmoidal nerve (V1).

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.